Dysarthria Detection – MFCC, wav2vec2, perceptually enhanced methods analysis + severity

Speech Signal Processing (EC5.408)

Spring 2026

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What is Dysarthria?

- A group of speech disorders resulting from disturbances in the muscular control over the speech mechanism.
- Results from damage to the CNS/PNS.
- The person knows what they want to say – but cannot control the muscles to say it clearly
- Commonly called slurred speech.

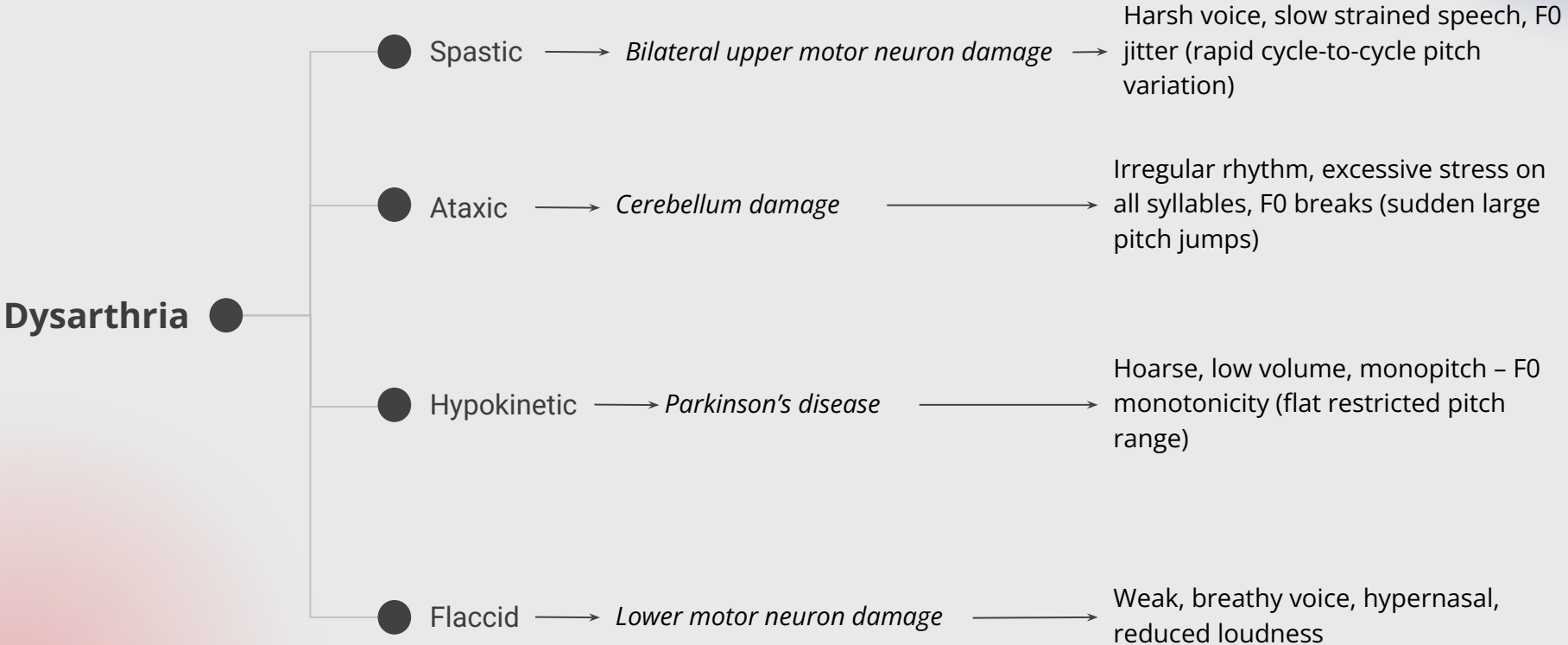
- Leads to disturbances in the articulation (tongue, lips), phonation (vocal cords), respiration (airflow), resonance (vocal tract and nasal cavity), and prosody (pitch, rhythm)

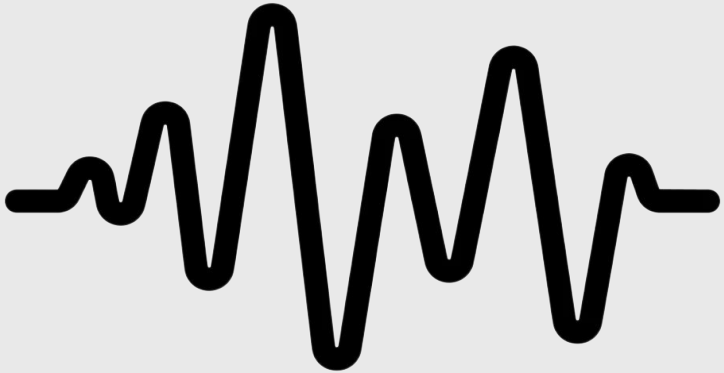
Causes can be multiple, to list a few:

- Stroke, Parkinson's, Cerebral Palsy
- ALS (Amyotrophic Lateral Sclerosis)
- Brain Injury, Muscular Dystrophy

Types of Dysarthria & Clinical Pitch Patterns

- Different neurological damage leads to different acoustic signatures.





01

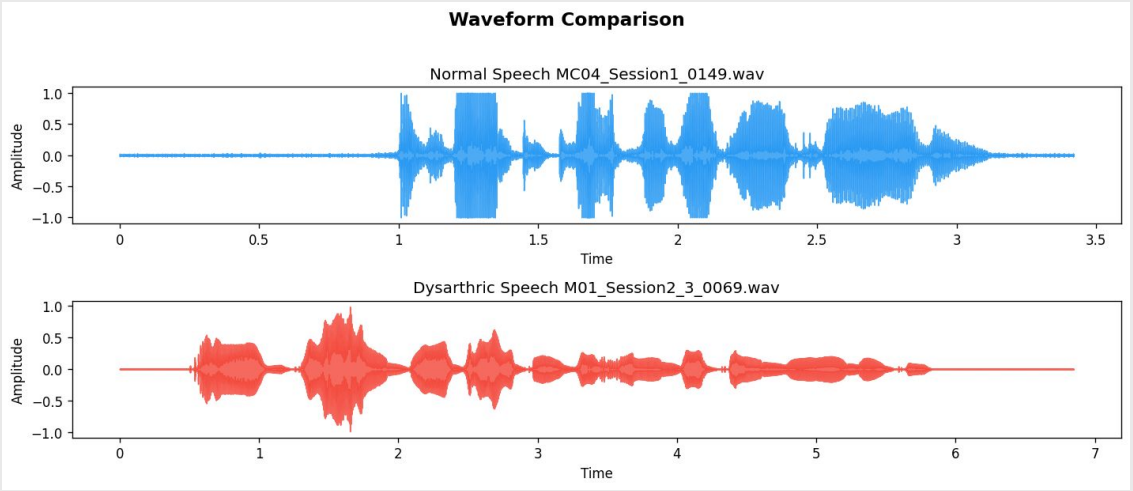
Visual Acoustic Analysis

Trying to get an intuitive understanding of how dysarthric speech differs from normal speech

Waveform Analysis

- Total duration (3.4 vs 6.8s)
  Speech duration (2 vs 4.5s)
- Speech rate is much slower, due to impaired motor control and slower articulatory movement.
- More frequent pauses and irregular spacing between segments – difficulty in coordinating respiration, phonation and articulation.
- NS - sharp distinct bursts of energy, clear syllable boundaries.
  DS - energy envelope melts into the other, unclear syllable boundaries (hence slurred speech!)
- Last segment in DS: “community” – breathy, low energy.

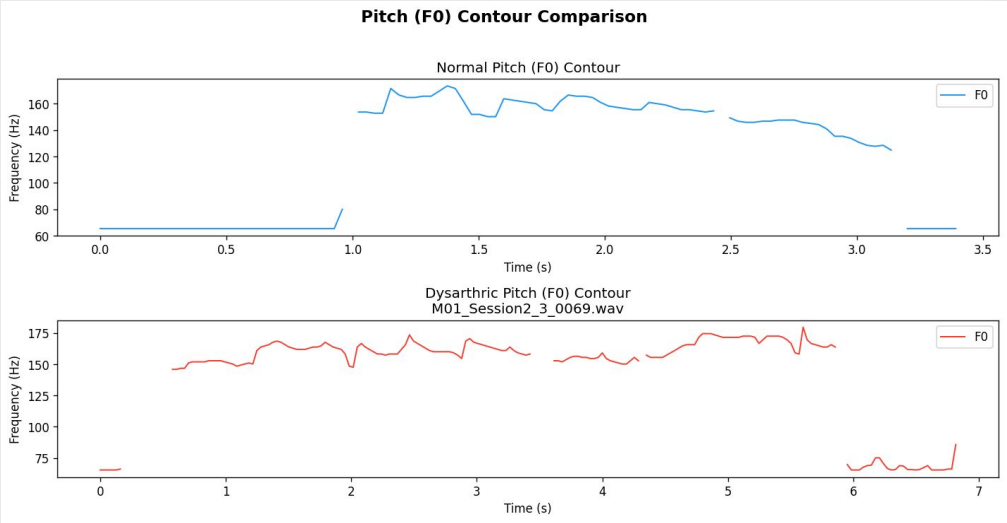
Sentence: “I tried to tell people in the community”



Default TORGO sampling rate: 16kHz

Pitch (F0) Contour

- YIN pitch detection algorithm (probabilistic version of it) in librosa used to determine the pitch contour.
- Healthy speech has a typical variance and cadence.
- Different types of dysarthria manifest as distinct disruptions in the F0 contour.
- DS typically falls into one of the two pathological classes – lower standard deviation due to vocal rigidity, or high std.dev due to erratic jumps/jitters.

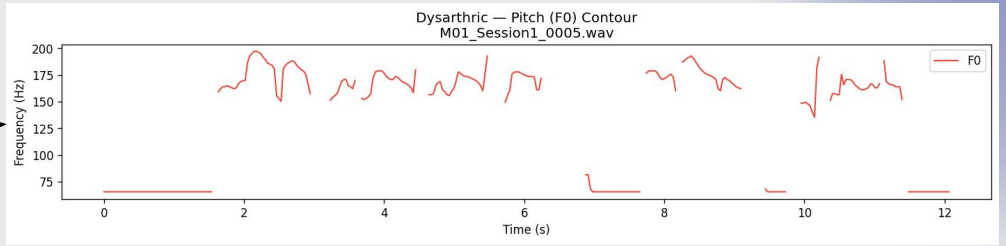


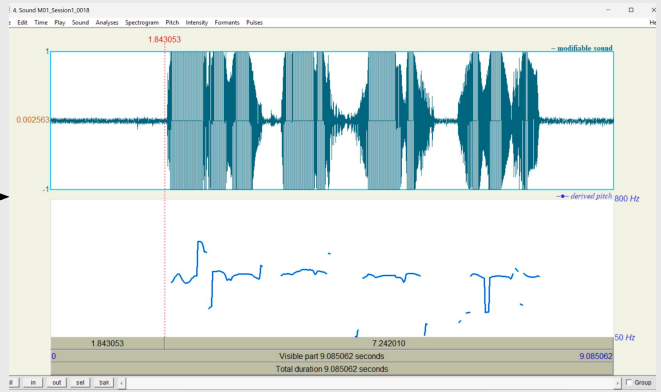
Pathological Analysis

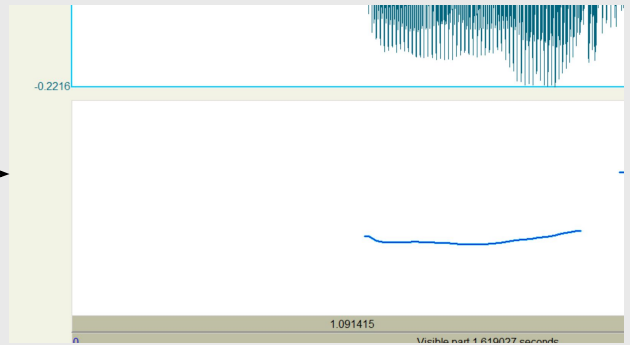
- F0 jitter (spastic pattern) – characterized by rapid, cycle-to-cycle variation in the fundamental freq. Lack of motor control over vocal fold vibration.

- F0 breaks (ataxic) – large abrupt shifts/interruptions.
  Folds fail to vibrate smoothly due to sudden changes in tension.

- F0 monotonicity (extrapyramidal) – abnormally flat, restricted pitch range lacking normal conversational prosody.
  Muscle rigidity and hypokinesia (Parkinson's)







Pathological Analysis

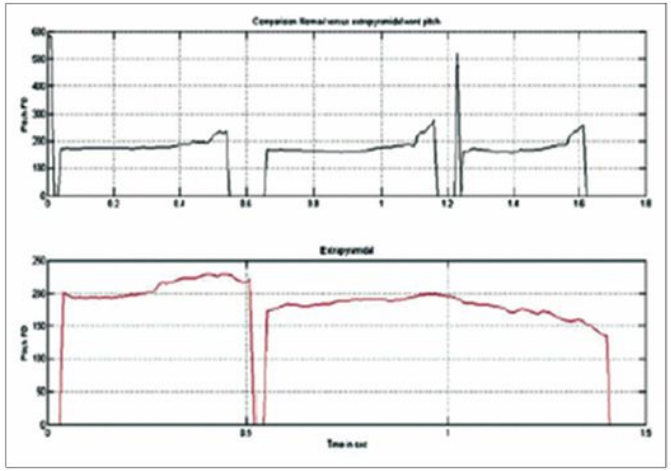


Figure 3: Comparison of pitch of normal speech with extrapyramidal speech. Demonstrates F0 monotonicity

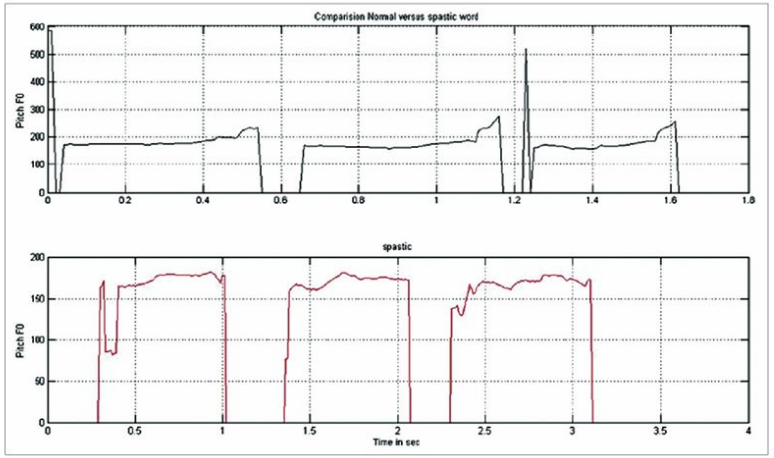


Figure 1: Comparison of pitch of normal speech with spastic speech. Demonstrates F0 jitter

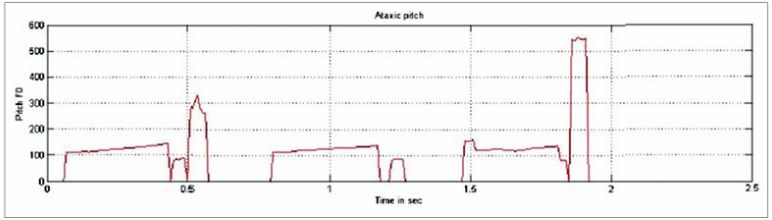
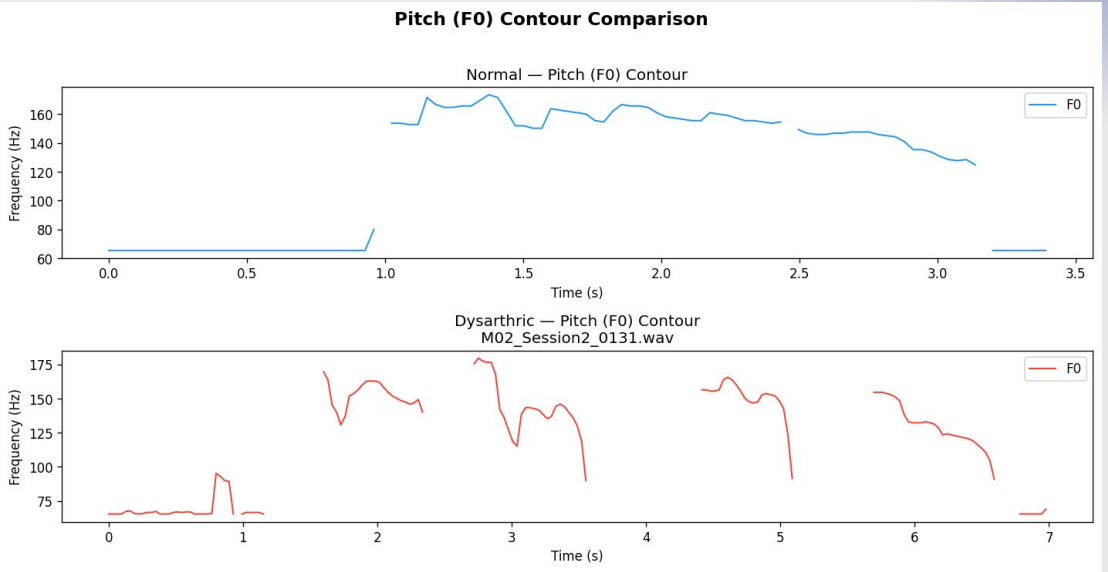


Figure 2: Comparison of pitch of normal speech with ataxic speech. Demonstrates F0 break

Source: [1]

Breathy Speech

- The pYIN algorithm shows unvoiced regions as white spaces.
- In the DS speech here, the unvoiced component corresponds to breathy speech.

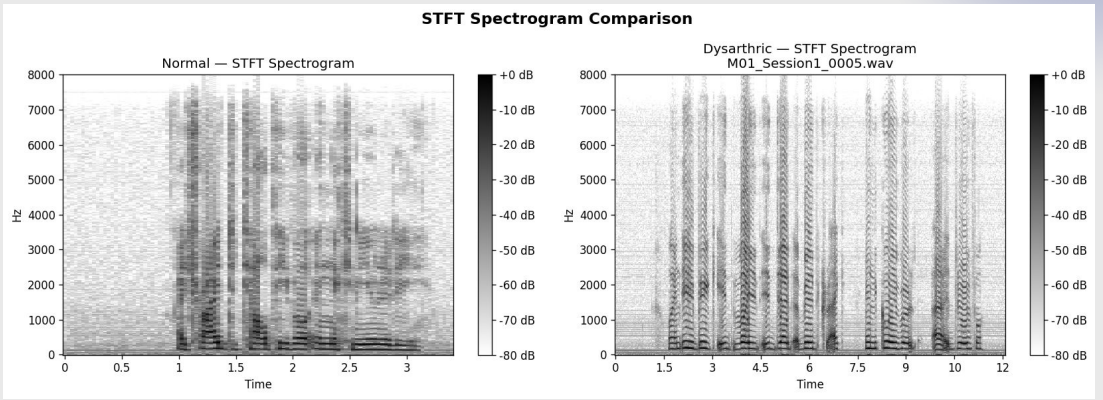


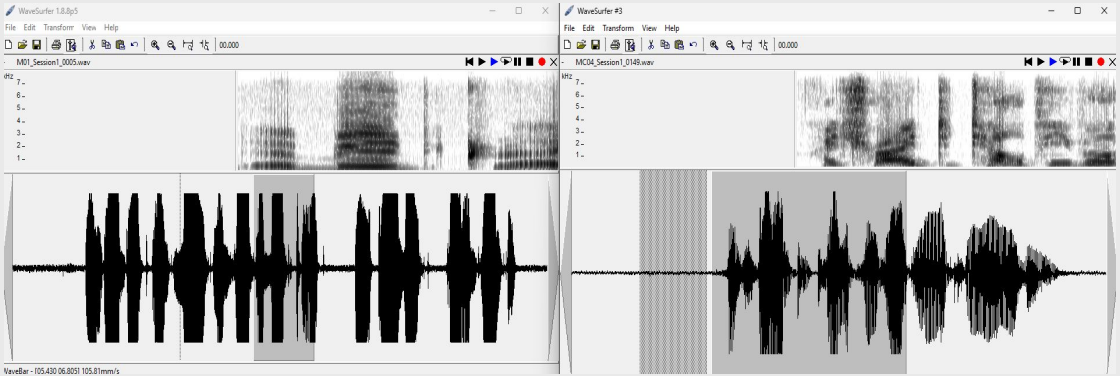
Spectrograms (STFT)

- Energy magnitudes of frequencies are more spread out across time – indicative of slow and slurred speech.
- Longer time duration.
  NS - Fast and dense segments
  DS - Slow and widely spaced segments.
- Frequency of pauses is more in DS.
- Formants (dark patches) are sharper and concentrated in NS, smeared and less distinct in DS.
  Reduced articulatory precision.
- In some breathy cases, we also observed high-frequency energy to appear more diffused.

Default TORGO sampling rate: 16kHz

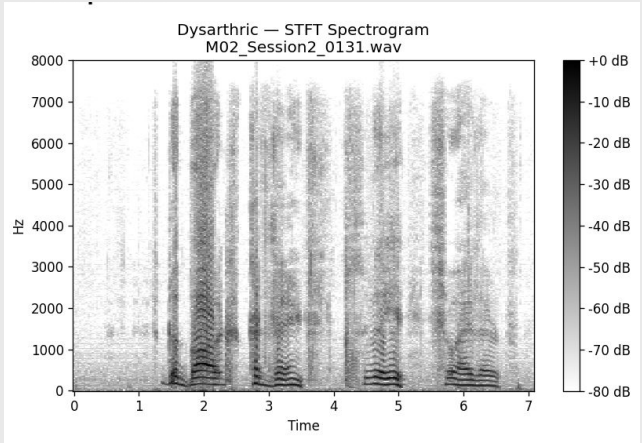
n_fft = 512, hop length = 32ms, window-length = 128ms

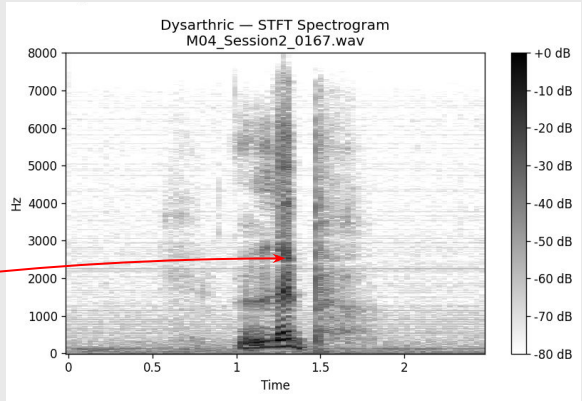




Breathy Speech - Spectrograms

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Post this stop, the speech is just breathy, no words spoken!

MFCCs

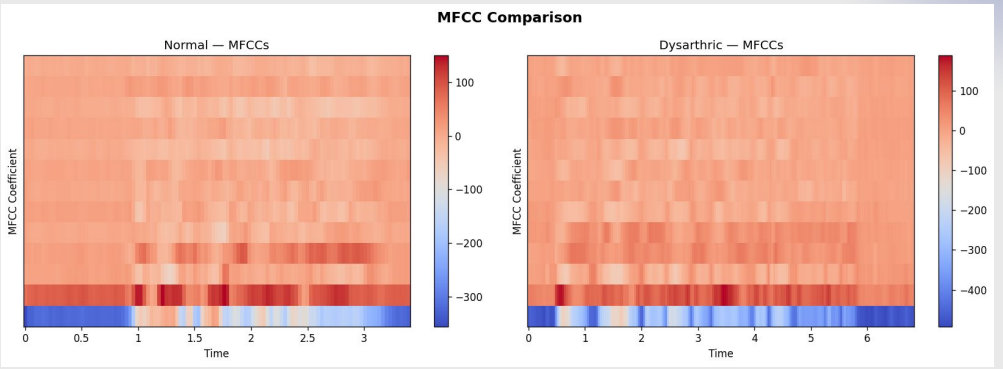
- Heatmap tracks the shape of VT changing with time.
  Lower ones representing broadly the VT shapes, higher ones about the finer spectral details.

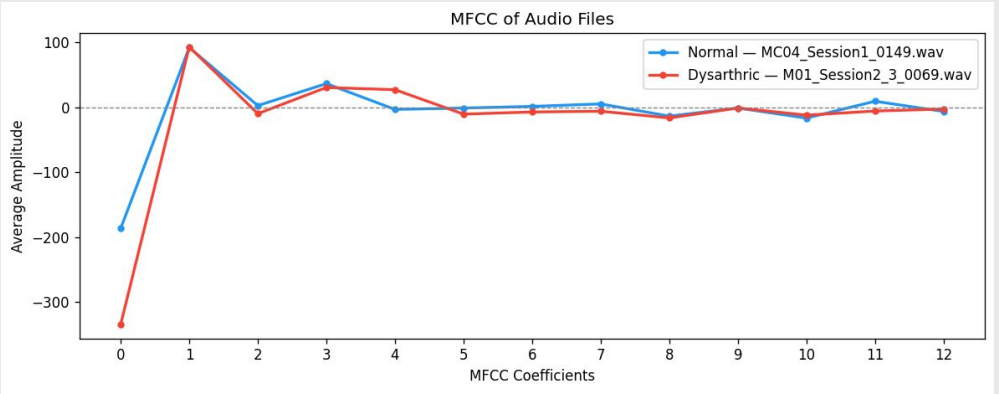
- The lowest coeff is about average energy, which is lower for DS, due to poor breath support and fading volume, esp. towards end of speech.

- Divergence in coeffs. at C4, abnormal baseline/restricted VT?
  Only lower two rows have more variations in DS – rigid posture of VT.

- DS have less concentrated regions in the heatmap (while in NS it is spread out) (even for the higher coeffs)

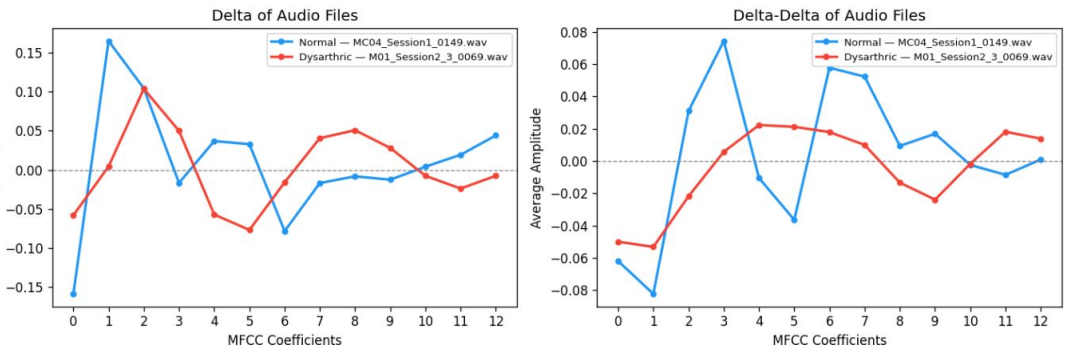
Default TORGO sampling rate: 16kHz, n_mels = 128, delta and delta2 included (39 features)
n_fft = 2048, hop length = 32ms, window-length = 128ms

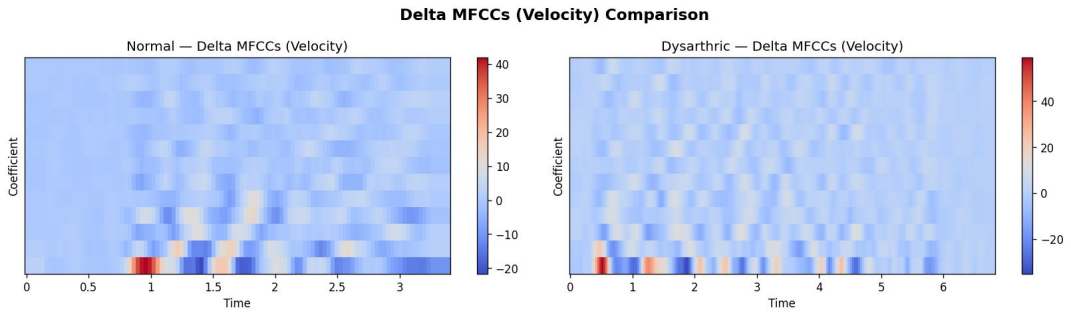


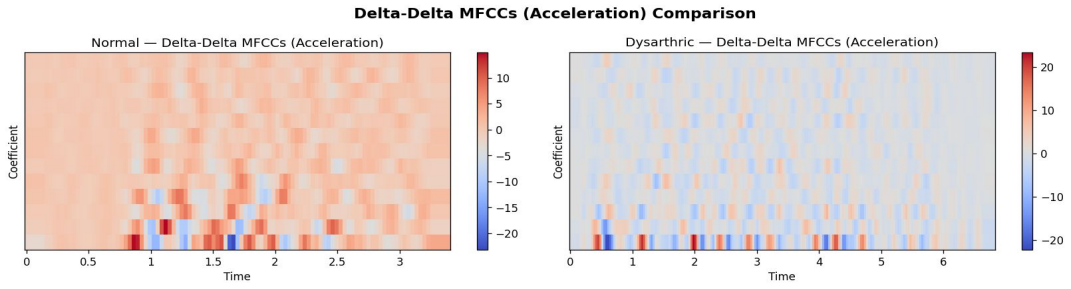


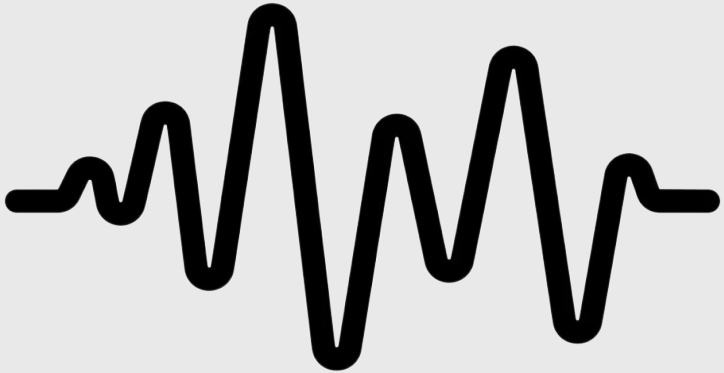
Delta-MFCCs

- In the delta MFCCs, the normal speech has sharper colors, showing more fluctuations and more rapid movement. Notice in DS, the variability is lesser in both delta and delta2.
- Delta2 map is significantly washed out for DS, while NS has pockets of acceleration.
- Dysarthric speakers cannot simply generate the motor acceleration required for crisp speech that easily.









02

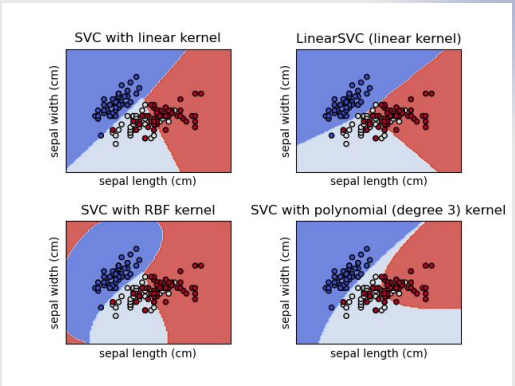
Training an SVM with MFCC

Running models to gain insights!

Running the SVM with different kernels

- Why the SVM? — Small dataset of ~2000 data-points for the reduced TORGO set. Speech involves non-linear in nature (frequency, phonetic content, acoustics), need a non-linear decision boundary for detection. Also, high-dim data.
- For each utterance, a 39-length feature vector was created (MFCC, delta, delta2), by taking a temporal average, so the data is fixed size for the SVM.
  Mean thus covers the average spectral shape and changes.
- A 78-dim vector was also created, with std-dev of each of the 39 features, since dysarthric speakers with restricted articulation have lesser variability.

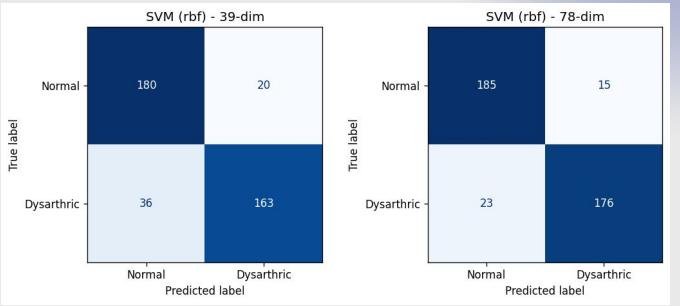
The delta and delta2 coefficients help partially recover some temporal dynamics after mean pooling!



- Trained with 3 kernels – RBF, poly, linear. Best kernel selected by 5-fold CV mean accuracy on training set.
- RBF outperformed in both feature sets, an expected result intuitively!

SVM results

Feature Set	Kernel	Train	Test	5-fold CV
39-dim (mean only)	RBF	0.921	0.860	0.857 ± 0.027
78-dim (mean + std)	RBF	0.968	0.905	0.910 ± 0.012

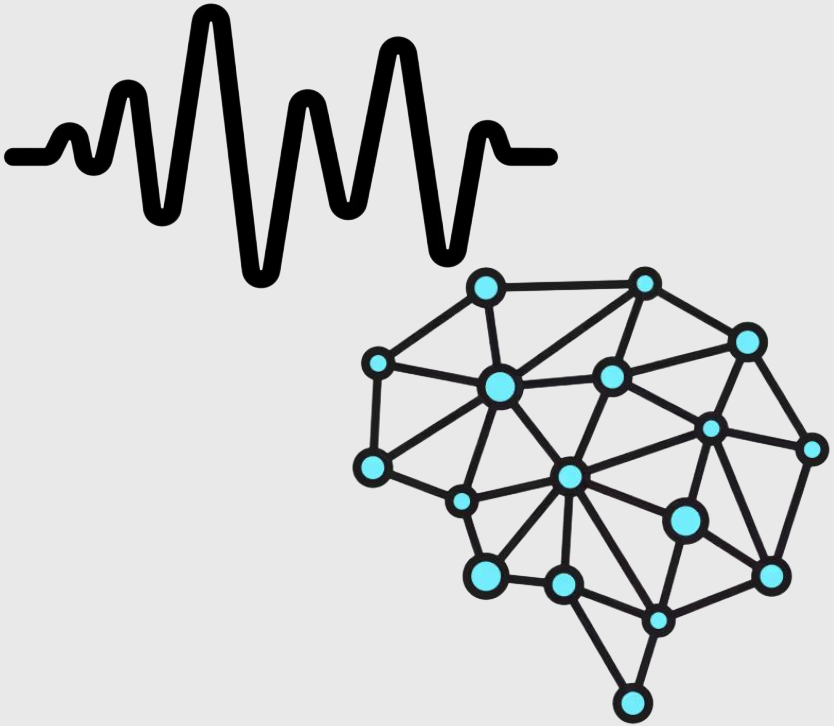


- Dysarthric speech is clinically heterogeneous (spastic, flaccid, and ataxic subtypes produce distinct MFCC distributions),
- This requires a non-linear decision boundary.
- Adding std features improved CV accuracy from 85.7% to 91.0%, confirming that articulatory variability carries discriminative information beyond the mean spectral shape alone.

Class	Precision	Recall	F1
Normal	0.83	0.90	0.87
Dysarthric	0.89	0.82	0.85
Macro-Average	0.86	0.86	0.86

The limitations could be due to mean-pooled features, since collapsing the time axis discards the sequential articulation dynamics that characterise dysarthric speech, such as irregular syllable timing and erratic phoneme transitions.

***These were simple runs to get familiar anyway, the main runs are shown in the next sections.*



03

Wav2Vec2 Analysis

Let’s look at the story hidden in the layers!

Why has DL come up in the recent years for DS detection?

- The speech intelligibility assessment is conventionally performed by speech-language pathologists in voice clinics using standard intelligibility tests (Kent et. al, 1989).
- Subjective intelligibility tests however, are **costly and laborious**, and subject to **pathologists' biases** because of familiarity with patients and their speech disorders (De Bodt et al., 2002a).

- The development of an objective method is important for the assessment of dysarthric speech. This involves (1) detection of DS, and (2) Severity classification.

- In recent years, automatic detection and severity level classification of dysarthria from speech has received considerable attention due to progress in signal processing as well as in machine learning (ML) and deep learning (DL). These models were originally pre-trained on a large quantity of unlabeled speech data and fine-tuned on a small labeled set for ASR.

- Therefore, the final layer of the pre-trained models have mainly learned speech representation that contain phoneme-related information. The learned speech representations from the lower layers of the network, however, contain information related to phone. These lower layers of the network can therefore be utilized as features in various downstream speech tasks, including the classification problems like we do here!

What is wav2vec2?

- Architecture consists of a multi-layer CNN encoder, a context network (12 transformer layer), and a quantization module.
- CNN encoder transfers speech segments of 20ms into latent representations (***Z***), fed to a feature projection layer to match the inner **768-dim** of the context network.

- Quantization module takes the cont. features ***Z*** and converts to discrete speech targets ***Q***.

- In pre-training, you force the context network to learn by masking (randomly selects a proportion of time steps, ***p***). Positional embeddings are added too. The network then guesses and generates its context representation ***C***.

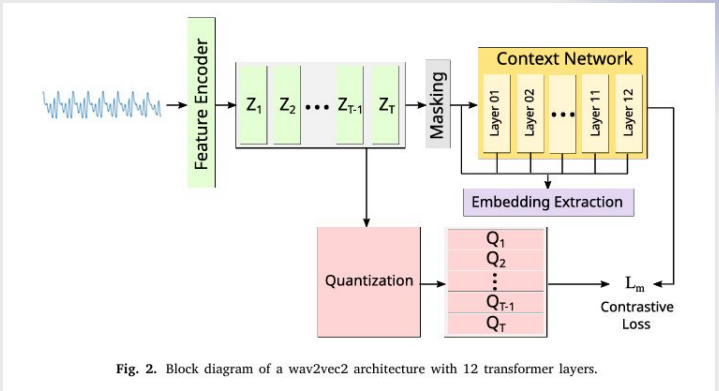


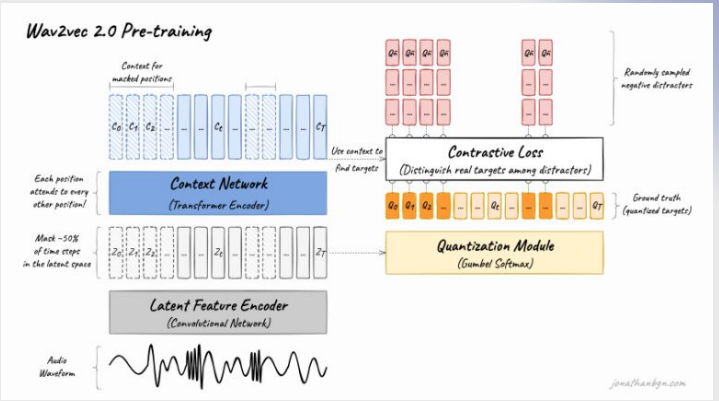
Fig. 2. Block diagram of a wav2vec2 architecture with 12 transformer layers.

- Contrastive loss objective function used. Model's ***C*** is compared against 101 (1 true, 100 false "distractors").

- These distractors are uniformly sampled from other masked steps with the same utterance to make it tougher!

Image source: [2]

What do the layers do?



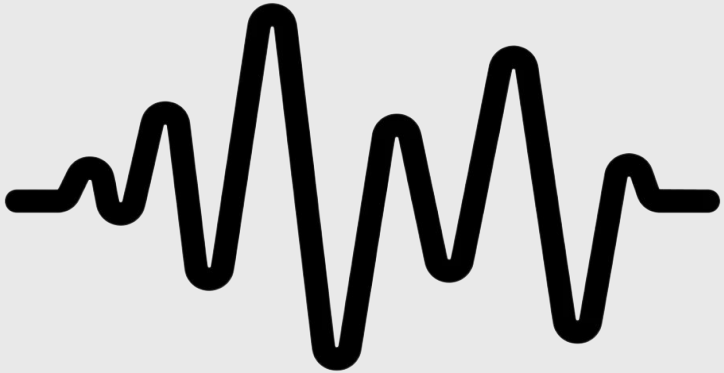
- Wav2vec2 are deep transformer networks fine-tuned for specific tasks like ASR.
  They thus process audio closer to the human brain, than classical models, starting with raw sounds and building up to language.
  Acoustic specialists (lower layers) and Linguistic specialists (upper layers)

- Learned speech representations from the initial/lower layers contain info. related to the phone. Also, raw acoustic properties like pitch and timbre. The model is just analyzing the physical sound waves, much like MFCC and other classical features.

- Final layers have learned representations containing phoneme-related info. As you move up, there's a shift from basic acoustic features to more complex linguistic info (syntax, semantics).

The model no longer looks at just the sound wave, but tries to interpret those sounds as words and sentences based on the data it was trained on!

Dysarthria primarily affects the raw acoustics and articulation (the *phones*), which the lower layers capture perfectly, atleast at the word level. Ofcourse, sentences would need more details, which the higher levels are better at.



04

Wav2Vec2 vs MFCC on UA-Speech

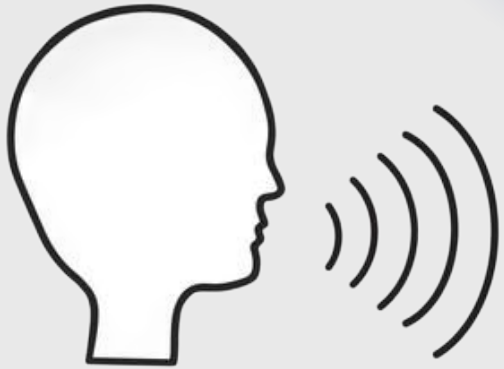
Using real datasets to gain insights!

About UA-Speech Dataset

- UA-Speech Dataset contains **765 isolated words** record across three blocks, by 15 dysarthric speakers diagnosed with cerebral palsy, and 13 healthy control speakers.
- Speakers were recorded using an 8-mic array at **16kHz**.

- Dysarthric speakers are grouped into four severity categories – very low, low, medium and high, baked on intelligibility ratings from 5 native listeners.

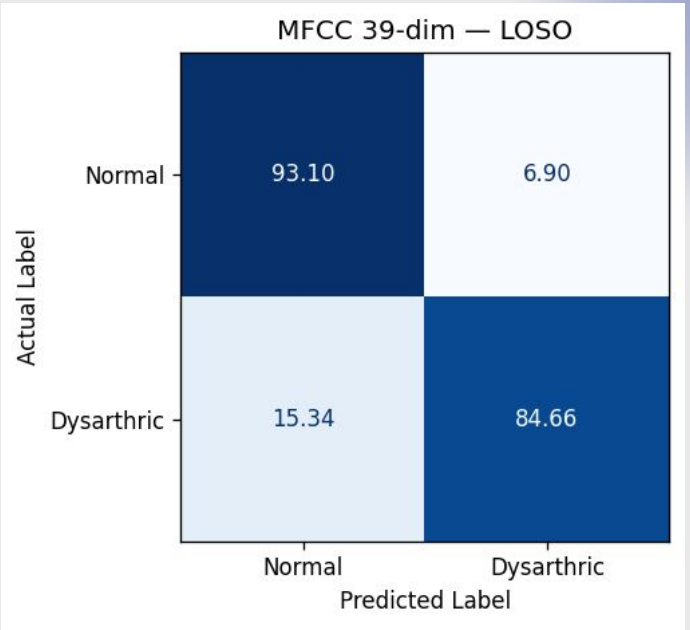
- For simplicity, and due to local compute, only one of the datasets (original) was used to analyze and get results.



MFCC + SVM Classifier

- Frame length = 25ms, hop = 5ms,
- n_mfcc=13, delta and delta2 included, so a 39-dim vector.
- Temporal pooling done like before.
- Balanced accuracy - 88.88%

  This sets a baseline to compare with wav2vec2.




```
LOSO accuracy - mean per fold:  88.58%
Balanced accuracy (aggregated): 88.88%

              precision    recall  f1-score

      Normal       0.84      0.93      0.88
  Dysarthric       0.93      0.85      0.89

    accuracy                           0.89
   macro avg       0.89      0.89      0.89
weighted avg       0.89      0.89      0.89
```

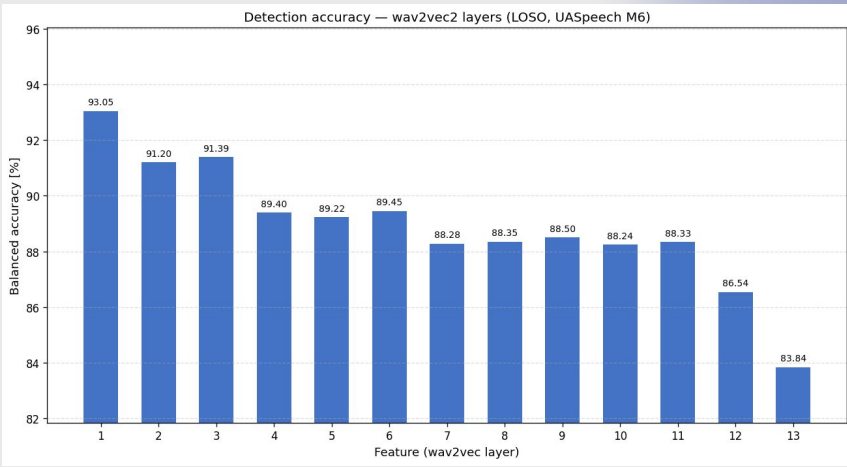
Source: [3]

Wav2vec2 - Layerwise Analysis

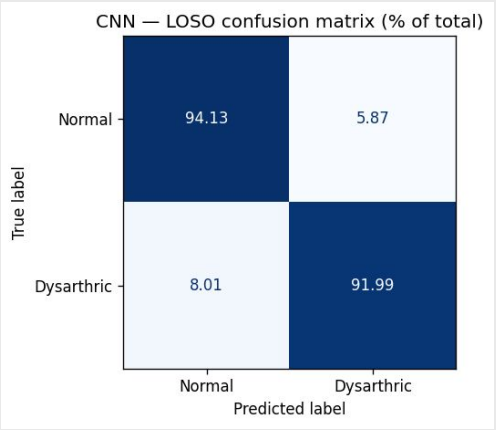
- The same RBF-SVM setup from MFCC baseline is used, per layer. By keeping the classifier constant and varying only the layer, any accuracy changes are attributable purely to the representation, not the classifier.

- The initial layers, have outperformed the MFCC. Wav2vec-1 showed an improvement of ~4% in accuracy.

- This finding indicates that the first embedding layer of the wav2vec model learns generic speech representations that are useful enough for distinguishing dysarthric speech from healthy speech.



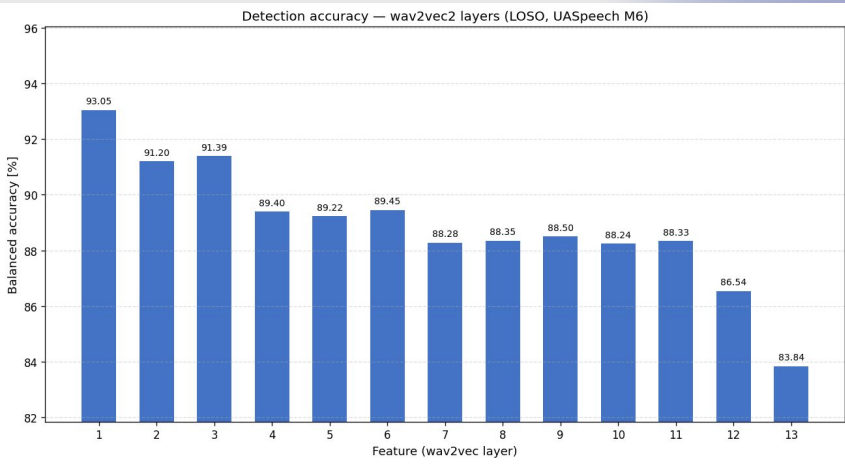
- The dataset contains only isolated word pronunciation – a single utterance type with no complex sentence-level prosody or connected speech dynamics. Dysarthric distortions in this context are primarily acoustic and phonetic; abnormal spectral shape, breathiness, reduced vowel space, all of which are captured strongly in the CNN projection layer which encodes local acoustic frames without much temporal context.

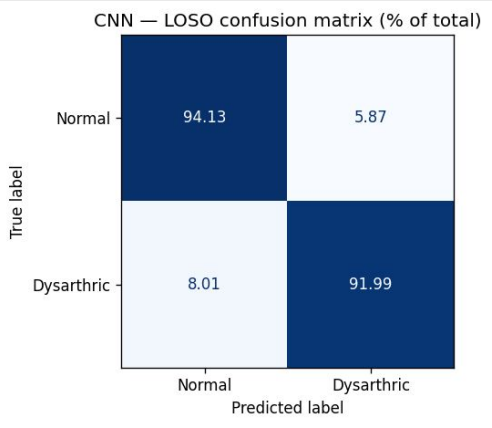


Source: [2,3]

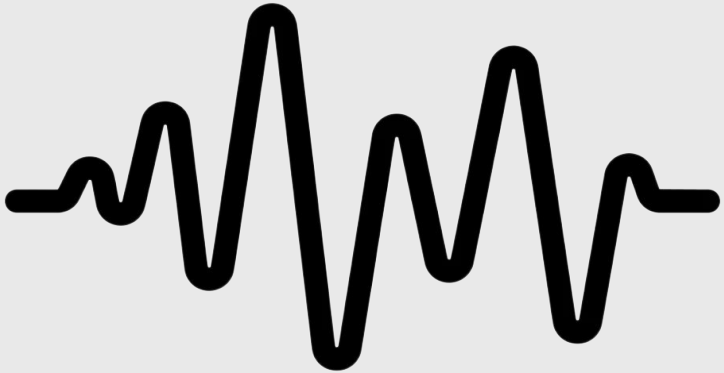
Analyzing Results

- The layer accuracy curve shows a falling trend. This indicates that the embedding layer learns generic representations (ofcourse, better than MFCC), that are useful for distinguishing DS from NS.
- The initial layers, cover basic phonetic elements, acoustic properties like pitch, timbre. These are crucial for identifying DS.
- Ds often manifests itself as atypical articulation and acoustic variation. As we progress from the initial layers to the final layers, there is a notable shift in focus from basic acoustic features to more complex linguistic information... these aspects are typically less linked to the specific characteristics of dysarthric speech!





Sources: [2,3]



05

Wav2Vec2 vs MFCC on TORGO

Using the more complex dataset!

Sources: [2]

About TORGO Dataset

- The TORGO database is a publicly available dysarthric speech dataset recorded from 8 dysarthric speakers (3 female, 5 male).
- Speech was recorded simultaneously using a head-mounted microphone and an array microphone, sampled at **16kHz**, in **mono** format.
- The dataset used in this project consists of approximately 500 .wav files per category across four folders: dysarthric male, dysarthric female, normal male, and normal female.
- A slightly more complex dataset than UA-Speech, in terms of speech info.
  Contains 3 categories of speech signals, non-words, words and sentences. Intuitively I thus expect the later lays to capture the features better.

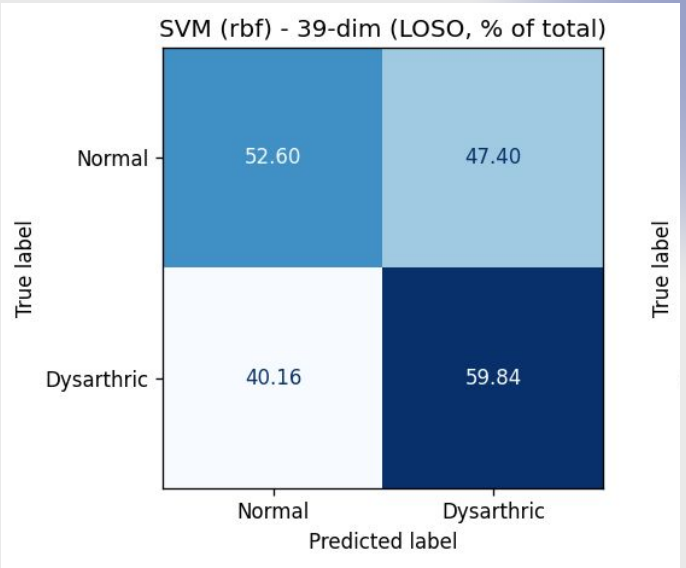


Speaker	Utterances	Label
F01	74	dysarthric
F03	292	dysarthric
F04	133	dysarthric
FC01	39	normal
FC02	230	normal
FC03	231	normal
M01	89	dysarthric
M02	108	dysarthric
M03	100	dysarthric
M04	76	dysarthric
M05	119	dysarthric
MC01	171	normal
MC02	80	normal
MC03	153	normal
MC04	96	normal

Total: 1991 utterances, 15 speakers
LOSO folds: 15

MFCC + SVM Classifier

- Frame length = 128ms, hop = 32ms, n_fft=2048 (default)
- n_mfcc=13, delta and delta2 included, so a 39-dim vector.
- Temporal pooling done like before.




```
=== 39-dim (mean only) ===
  SVM (rbf    )  LOSO = 0.652 +/- 0.309
  SVM (poly   )  LOSO = 0.567 +/- 0.317
  SVM (linear)  LOSO = 0.602 +/- 0.293
  → Best kernel: rbf

  Classification Report (LOSO, 39-dim (mean only), rbf):
              precision    recall  f1-score   support

      Normal       0.57      0.53      0.55      1000
  Dysarthric       0.56      0.60      0.58       991

    accuracy                           0.56      1991
   macro avg       0.56      0.56      0.56      1991
weighted avg       0.56      0.56      0.56      1991
```

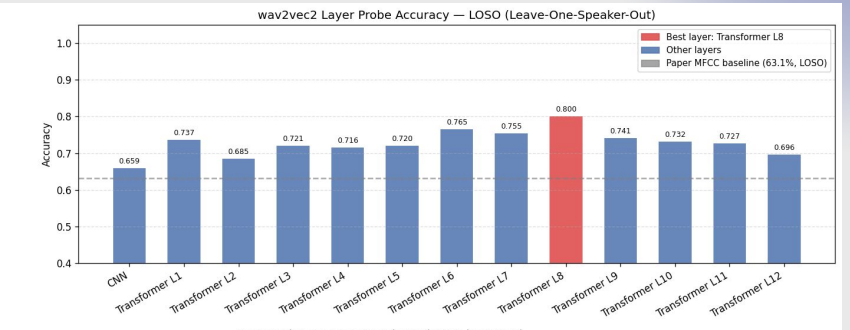
RBF again outperforms the other kernels.

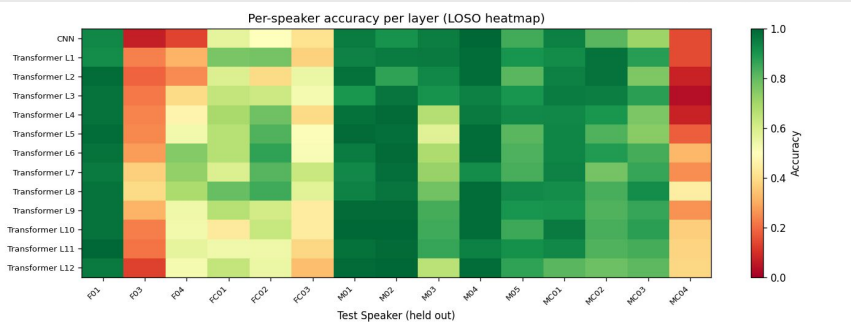
Given the complexity of the dataset, and LOSO-learning, the drop in performance is expected.

Source: [3]

Wav2vec2 - Layerwise Analysis

- The layerwise analysis is all about the question “at which depth of the network does DS become most linearly separable from NS”
- The pipeline is same as that of the MFCC, to keep experimenting conditions same, except that linear kernel was used.
- Layer-8 output performs the best. The layer accuracy curve shows a roughly rising trend from the CNN layer through Transformer L8, followed by a falling trend toward L12.
- Can be explained by the nature of the TORGO dataset – words, non-words and sentences, thus making it more complex than simple isolated word databases.
- A richer or more complex database (TORGO) might benefit more from the deeper, more abstract representations in later layers. On the other hand, a simpler database (UA-Speech) might be adequately represented by the initial layers. [2]

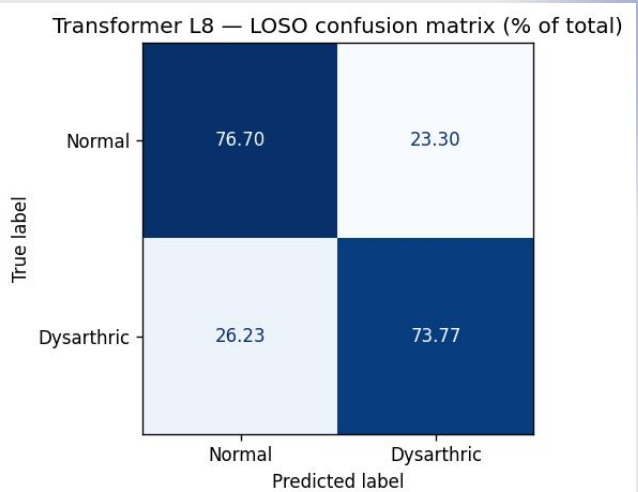




Source: [2]

Analyzing Results

- The confusion matrix results are better too, showing that the wav2vec-8 just learns and captures features better than what MFCC does.
- The middle layers build phonetic and coarticulatory representations, viz. how sounds transition into each other, how phoneme boundaries are shaped, how the vocal tract moves across a sequence.
- Capturing these dynamics helps in sentence classification, which is lost in temporal pooling in MFCCs. wav2vec2's self-attention mechanism attends over all frames simultaneously, allowing the model to encode how articulation unfolds over time, which is precisely what dysarthria disrupts. (Each layer adds a capability MFCC alone cannot provide)



Also note, these models may have been inadvertently exposed to dysarthric speech during pre-training, since large corpora inevitably include speakers with undiagnosed pathologies.

This might partly explain why wav2vec2's representations, trained on healthy speech, still generalise so well to dysarthric detection.

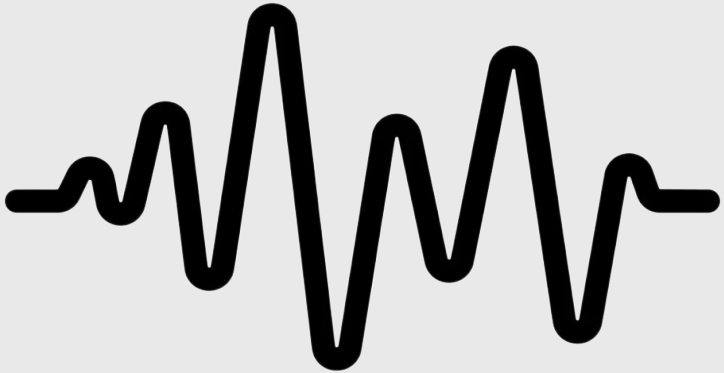
Sources: [2,3]

So, what's next?

- Implement the paper [7] (*Gurugubelli, K., Vuppala, A. K.* ***Perceptually Enhanced Single Frequency Filtering for Dysarthric Speech Detection and Intelligibility Assessment***) and analyze with MFCC/wave2vec2.
- Implement a severity classifier for dysarthia and replicating the results in the paper: WAV2VEC BASED DETECTION AND SEVERITY LEVEL CLASSIFICATION OF DYSARTHRIA FROM SPEECH

Open to feedback and changes here!

*If time permits, IIIT acads not too kind with it:P

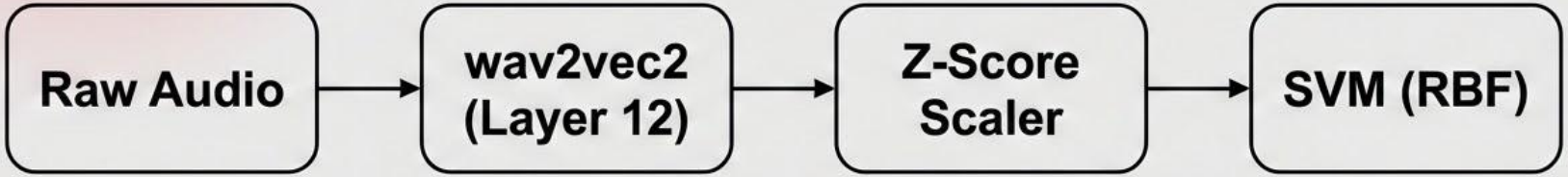


06

Dysarthria Severity Classification

- Fulfilling the mid-evaluation proposal to move beyond binary detection.
- Objective: Multi-class severity assessment on the TORGO database.
- Classes: None, Very Low, Low, and Medium.
- The Challenge: Extreme class imbalance heavily skewed toward normal speech, creating a high risk of model overfitting.

The Hybrid Pipeline (wav2vec2 + SVM)



Feature Extraction & Classification Strategy

- **Feature Extractor**: wav2vec2-base (Layer 12).
  - **Why Layer 12?** It captures the highest level of linguistic and phonetic contextualization, which correlates directly with clinical speech intelligibility.
- **Normalization**: Strict Z-score scaling to prevent high-magnitude acoustic dimensions from distorting the classifier.
- **Classifier**: Support Vector Machine (RBF Kernel).
- Implemented balanced class weights to force the model to mathematically respect the minority severity classes (Very Low, Low).

Initial Scores

```
Overall Accuracy: 73.33%

              precision    recall  f1-score   support

        None       0.75      0.99      0.85      2196
    Very Low       0.00      0.00      0.00       514
         Low       0.00      0.00      0.00       165
      Medium       0.65      0.59      0.62       436

    accuracy                           0.73      3311
   macro avg       0.35      0.39      0.37      3311
weighted avg       0.58      0.73      0.64      3311
```

Validation & Overall Performance

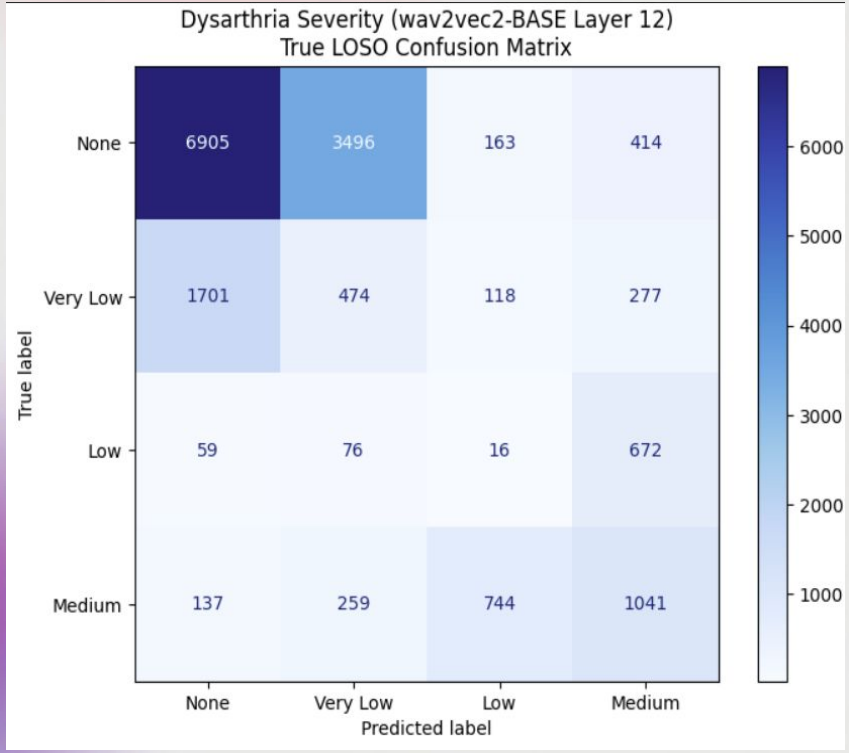
```
Running Leave-One-Speaker-Out (LOSO) Evaluation...
Please be patient! SVM training takes a few minutes per speaker.

Tested on Speaker F01 | Accuracy: 4.2%
Tested on Speaker F03 | Accuracy: 14.4%
Tested on Speaker F04 | Accuracy: 35.5%
Tested on Speaker FC01 | Accuracy: 64.2%
Tested on Speaker FC02 | Accuracy: 67.7%
Tested on Speaker FC03 | Accuracy: 58.0%
Tested on Speaker M01 | Accuracy: 52.2%
Tested on Speaker M02 | Accuracy: 55.1%
Tested on Speaker M03 | Accuracy: 9.6%
Tested on Speaker M04 | Accuracy: 34.2%
Tested on Speaker M05 | Accuracy: 1.0%
Tested on Speaker MC01 | Accuracy: 71.6%
Tested on Speaker MC02 | Accuracy: 49.1%
Tested on Speaker MC03 | Accuracy: 73.5%
Tested on Speaker MC04 | Accuracy: 49.1%

====================================
Final True LOSO Accuracy: 42.64%
====================================
```

- **The Data Leakage Trap:** Standard random splits yielded artificially inflated accuracies (~73%) due to the model recognizing speaker identity rather than disease severity.
- **The Fix**: Enforced strict Leave-One-Speaker-Out (LOSO) cross-validation.
- **Final LOSO Macro-Accuracy**: 42.64%
- **Literature Comparison**: Matches and slightly exceeds the baseline set by Javanmardi et al. (2024), who reported 42.12% using traditional eGeMAPS features.

Confusion Matrix Analysis – The Clinical Reality

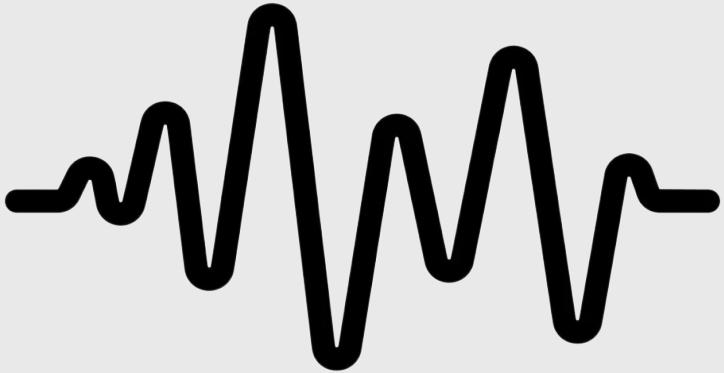


Analyzing the Confusion Matrix

- **The Clinical Gray Area**: 3,496 'Healthy' files were misclassified as 'Very Low'. Differentiating healthy speech nuances (accents, mild slurs) from subtle neurological degradation is highly ambiguous even for clinicians.
- **The 'Low' Severity Collapse**: The model almost entirely failed the ‘Low’ class (only 16 correct, 672 misclassified as ‘Medium’).
- **Cause - Dataset Starvation**: The TORGO database only contains two ‘Low’ severity speakers. Under LOSO, the model lacks sufficient diversity to learn the distinct C_{low} acoustic boundary.

Final Takeaways

- Deep learning feature extractors (wav2vec2) effectively eliminate the need for manual acoustic feature engineering (MFCCs).
- Addressing data leakage (via LOSO) and class imbalance is absolutely critical in small medical speech datasets.
- Future improvements rely heavily on mitigating dataset starvation (gathering more diverse pathological samples) or exploring advanced transformer fine-tuning.



07

Perceptually Enhanced SFF

A better time-frequency representation?

About PE-SFCC

shift – 5ms, 39-dim feature vector

- Stands for Perceptually Enhanced Single Frequency Cepstral Coefficients.
- Motivation: Standard MFCC uses block processing, since it averages the spectrum over the window length into consideration.
- This (MFCC) can smear out the rapid, irregular acoustic events, which are seen typically in Dysarthric Speech.
- You pass the speech through a single-pole complex bandpass filter bank, producing an instantaneous time-frequency distribution, and then you refine it using a series of auditory perceptual operators.

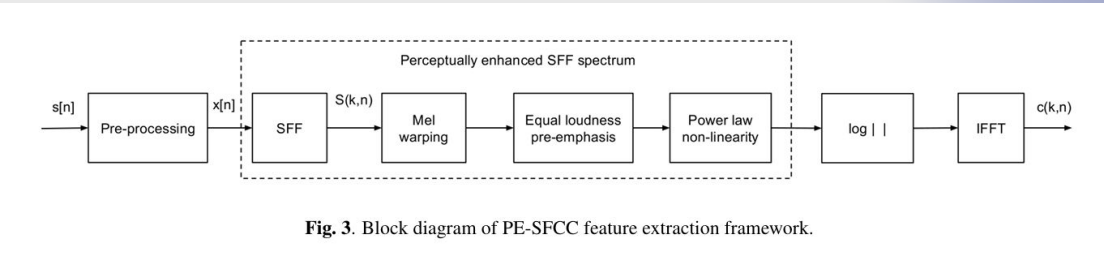


Fig. 3. Block diagram of PE-SFCC feature extraction framework.

sr = 8kHz, a = 0.98, range = 100-4000Hz, res = 20Hz — i.e. 196 filters, mel bands = 26, CCs = 13

Source: [7]

SFF

- First method in the pipeline, to generate a better, time-frequency distribution.
- SFF is a signal processing technique to extract the strength of a signal at a specific frequency of interest, but providing a higher spectro-temporal resolution.
- You shift the signal in freq. Domain and then apply a single-pole LPF.
- Since an LPF tracks freq. near 0Hz, you shift the filter using a complex sinusoid, to have a complex BPF tuned exactly to that frequency.
- You now pass the speech x[n] into this filter, and to get the loudness (due to real and imag. parts), you calculate the magnitude (envelope) of the complex output at that time instant.

$$H(z) = \frac{1}{1 - az^{-1}}.$$

Pole radius

$$H_k(z) = \frac{1}{1 - a_k z^{-1}} \rightarrow a_k = ae^{-jw_k}$$

```
y_k[n] = a_k · y_k[n-1] + x[n]
```

$$m_k[n] = \sqrt{y_{kR}^2[n] + y_{kI}^2[n]}$$

Repeat this tedious math for hundreds of different frequencies, you stack the 1D envelopes together and make the 2D spectrogram!

Source: [7]

Pole radius

$$BW = \cos^{-1}\left(\frac{4a - a^2 - 1}{2a}\right)$$

- Time-freq. resolution of the SFF is governed by the pole radius, since it controls the 3dB bandwidth of the filter.
- Top to bottom in this image: a=0.85-0.995, you can observe the filter trade off its time res. for freq. res.
- For the smaller values, the filter is small (i.e. wider bw), less selective, hence worse freq. Resolution. (note that formants cannot be seen at all)
- As ‘a’ increases, filter is more selective, and formant bands are seen, but the time events (vertical) start to blur.
- Sweet spot found in paper: pole radius 0.9875 — formants are clearly highlighted, but there is still enough time resolution to capture sudden, abnormal acoustic events. (formants need a high freq. res so this makes sense).

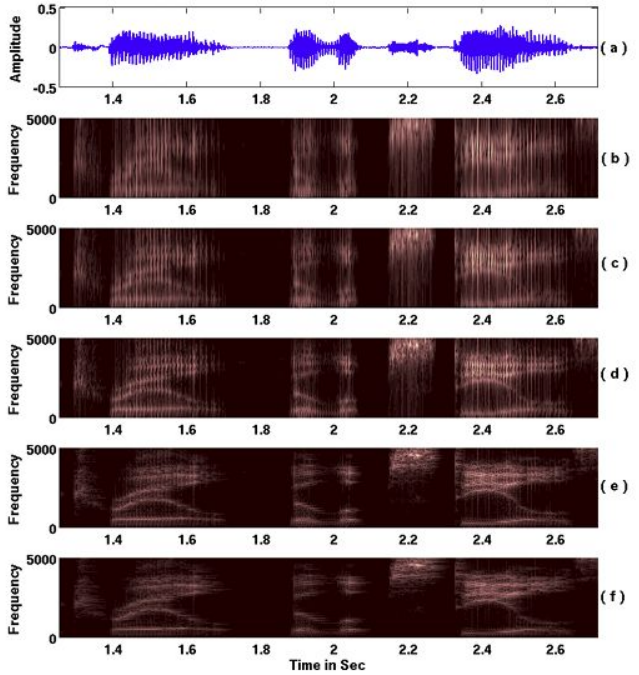


Fig. 2. Time-frequency representation of speech signal with different resolutions. (a) Speech signal. (b)-(f) SFF-TF representations of speech for a = 0.85, 0.90, 0.95, 0.99, and 0.995 respectively.

Source: [7]

Perceptual Operators

- The SFF gave us a high-res freq-time mapping, OK. But humans perceive audio differently! You now warp the map through a series of filters which mimics the features of the human ear, giving the PE-SFCC!

- **Mel warping**: Triangular mel-filterbanks applied to 196 SFF freq. bands reducing them to 26 mel bands.

  Equal distances on the Mel scale correspond to equal perceived pitch differences. (human ear has finer resolution at low frequencies and coarser resolution at high frequencies) — formants live in low freq!

- **Equal loudness pre-emph**: Equal loudness contour (Hermansky 1990, used in PLP analysis). the human ear is not equally sensitive to all frequencies at the same physical intensity.

  Around 3-4kHz the ear is most sensitive — a quiet sound there is heard clearly.
  Below 200Hz and above 6kHz, the ear needs much more physical energy to perceive the same loudness.

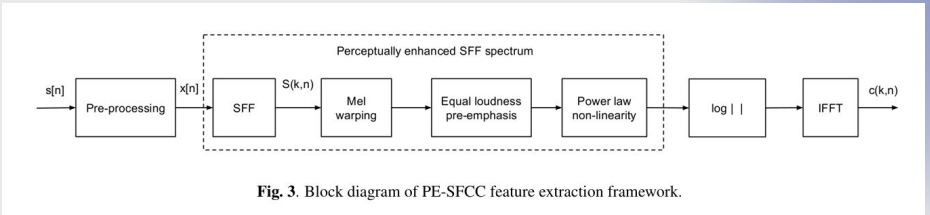


Fig. 3. Block diagram of PE-SFCC feature extraction framework.

You thus emphasise on freq. regions humans actually rely on to understand speech!

Perceptual Operators

- **Power Law Compression**: Note that the relationship between physical sound intensity and perceived loudness is non-linear.
  Doubling the physical energy does not double perceived loudness.

  Stevens' power law captures this. The exponent 1/5 (PNCC by Kim & Stern 2016) compresses the dynamic range of the spectrum to match how humans perceive differences in loudness.

- **Log + IFFT**: Same cepstral analysis as MFCC. The logarithm converts multiplication into addition, separating the vocal tract filter response from the excitation source. IFFT gives the cepstrum.

  The first few coefficients capture the slow spectral envelope (vocal tract shape), higher coefficients capture fine harmonic structure (source).

  Liftering keeps only the first 13 coefficients, retaining the vocal tract information and discarding pitch-related detail. Ofcourse adding delta and delta2 helps, as seen before.

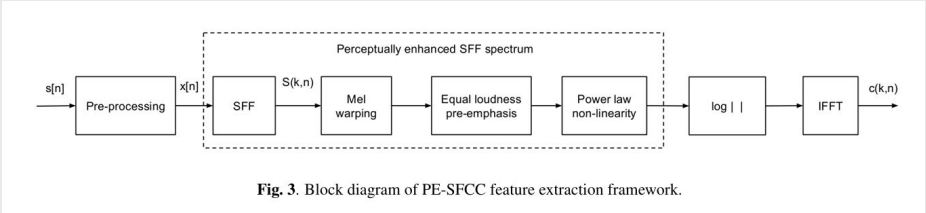


Fig. 3. Block diagram of PE-SFCC feature extraction framework.

Better than MFCC?

- The core argument is block processing versus instantaneous tracking.
- MFCC uses STFT with a fixed window. Every sample inside the window contribute to an averaged spectrum. But time and frequency thus come up to be a trade-off.
- SFF, instead of windowing, runs many (196) separate single-pole IIR filters in parallel, one per freq.
  The output updates every sample, since there is no window. The filter can thus track the energy at every freq. instantaneously through time.
- And adding the human-like features for perception, would definitely help. (like modelling the ear's sensitivity, and loudness perception instead of energy)

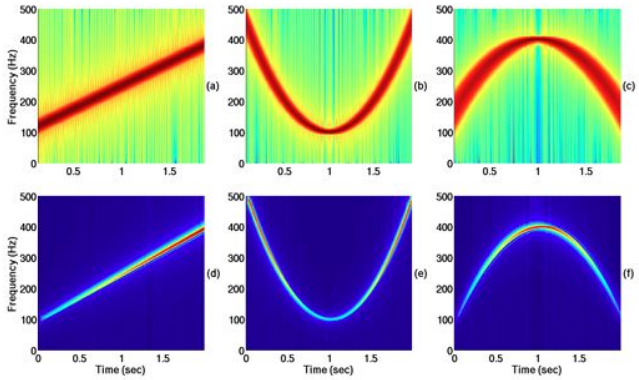


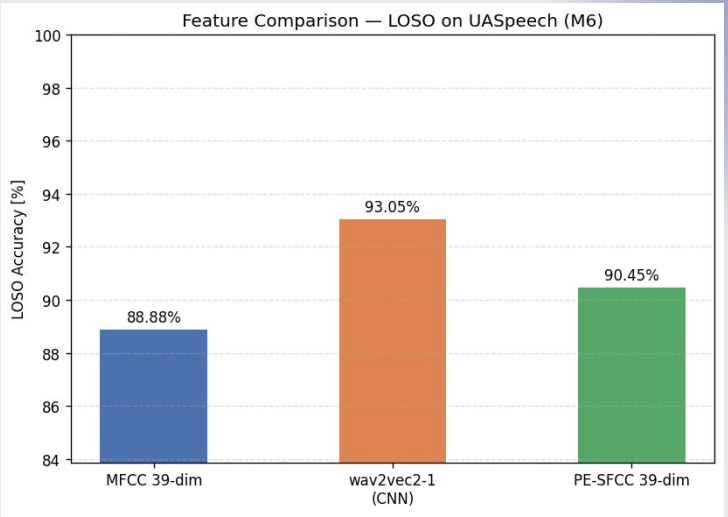
Fig. 1. Time-frequency representation of synthesized linear, quadratic, and convex chirp signals. Short time Fourier transform (Top row: (a)-(c)). Single frequency filtering (Bottom row: (d)-(f))

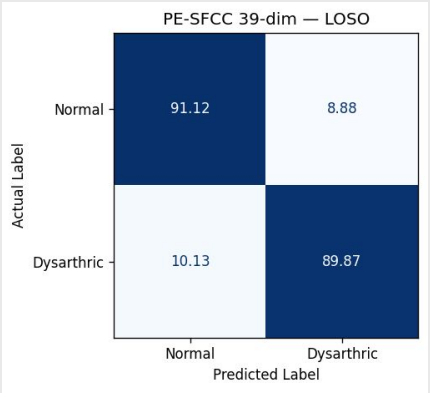
Frequency spread is less in case of SFF-TFR as compared to STFT representation. A better time-frequency localization can be observed in SFF-TFR. Since SFF is an instantaneous measurement, there is almost no freq. Spread!

Due to the better localization, If a dysarthric speaker has a sudden, rapid fluctuation in their pitch, STFT will blur it into a smudge, but SFF will track it as a sharp, highly localized curve!

Results on UA-Speech

- Performs better than MFCC as expected intuitively. LOSO accuracy – 90.45%, precision and recall ~0.90
- Thus, SFF's instantaneous tracking captures info that block-based MFCC misses. The better time-freq representation does make a difference.
- Cannot beat wav2vec2, because we still reduce the utterance to 39 coeffs. Maybe wav2vec2 encodes the signal in a way that makes DS and NS better linearly separable.
- The gap can also be due to a classifier, since an i-vector+PLDA used in the paper, is designed for speaker-independent classification.





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- [7] Gurugubelli, K., & Vuppala, A. K. **Perceptually Enhanced Single Frequency Filtering for Dysarthric Speech Detection and Intelligibility Assessment.**

... and a few more

Acknowledgments

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Thanks!